



Competitive fusion in multimodal networks for enhanced salient object detection

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Abstract

Salient object detection (SOD) aims to identify the most visually compelling regions in images, playing a crucial role in computer vision tasks. Traditional RGB-D/T multimodal interaction methods such as Hypersor with multiplicative operations, FasterSal via concatenation, and SACNet using equal interaction attention typically rely on equal interaction mechanisms and convolutional computations, exhibiting limitations in dynamically and adaptively capturing and representing critical discriminative features. For salient object detection, this paper introduces MC2FNet, a multi-scale cross-modal competitive fusion network for RGB and depth/thermal data, designed to achieve efficient intra-modal, cross-modal, and multi-scale competitive fusion. MC2FNet employs a hybrid attentional approach to reduce computational overhead and enhance feature extraction. Experimental results on RGB-T and RGB-D datasets demonstrate that MC2FNet achieves state-of-the-art performance, outperforming existing methods in accuracy and efficiency. This work advances the field by providing a novel framework for robust multimodal salient object detection. The code will be available at <https://github.com/liangjiaxiaoqi/MC2FNet>.

Keywords Salient object detection · multimodal competitive fusion · Group aggregation and re-embedding · Spatial synergistic suppression–enhancement

1 Introduction

The goal of salient object detection (SOD) is to identify the most visually striking regions in an image. It plays a fundamental role in various computer vision tasks, including target tracking [1], image understanding [2], medical image segmentation [3], and camouflaged object detection [4]. Moreover, with the rapid advancement of automation, SOD is now widely applied in robotic visual perception [5], UAV reconnaissance [6], saliency-guided ship tracking [1], and other related fields.

Traditional unimodal SOD networks are highly vulnerable to environmental interference, whereas multimodal SOD mitigates this issue through feature complementarity. However, most existing methods employ simplistic fusion strategies, including direct summation [7], multiplication [8], splicing [9], and equal interaction [10]. These approaches overlook the intrinsic limitations of multimodal features, hindering their ability to handle complex scenarios effectively. As shown in Fig. 1, the FasterSal[11] model, which employs directly spliced multimodal features, fails due to inadequate depth feature representation when a pigeon stands on a tree branch of similar depth. Similarly, the SACNet [12]

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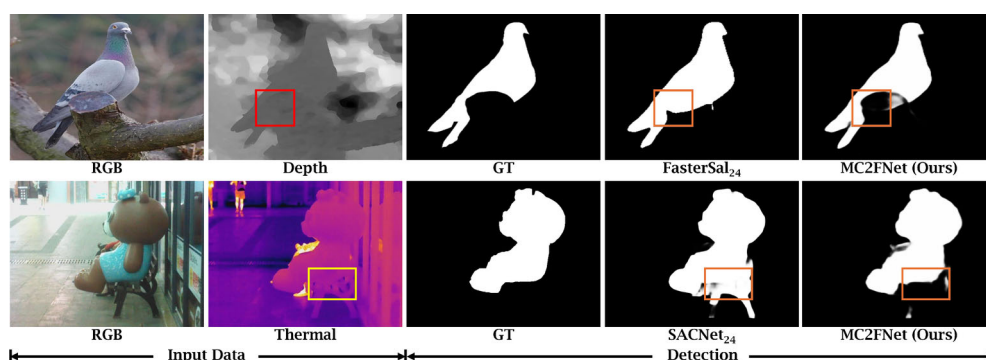


Fig. 1 In RGB-D/T salient object detection, the information representations of interference and target objects at the same depth or thermal are extremely similar, such as those of the red and yellow boxes in the figure, which may cause false detection. However, our proposed competitive fusion approach modeling the spatial synergistic

suppression–enhancement mechanism in visual cognition effectively solves this problem. Specifically, Columns 1 and 2 display input data from different modalities, Column 3 shows the ground truth (GT), Column 4 presents failure cases of existing methods, and Column 5 illustrates the detection results of our method

model, which utilizes equivalent interaction fusion, fails to detect a bear when its thermal features closely resemble those of a nearby chair. These examples illustrate that effective multimodal feature fusion in complex environments necessitates modal balance and dynamically adaptive fusion. Furthermore, with the rapid development of Transformer, attention mechanisms have been widely applied in various fields such as image processing [13], point clouds [14], 3D reconstruction [15], image enhancement [16], and wearable device-based human action recognition [17]. In the domain of multimodal salient object detection, cross-modal fusion leveraging attention mechanisms enables the perception of global information and the capture of long-range cues essential for recognition. However, traditional attention mechanisms face the challenge of computational complexity increasing quadratically with input sequence length.

To improve the accuracy of multimodal salient object detection (SOD) and address the issue of detection failure caused by target-background depth/thermal consistency or color-texture similarity (as shown in Fig. 1), we propose an innovative multimodal SOD network, MC2FNet. This method simplifies the attention mechanism, overcomes the limitations of convolutional computations, and enables long-range dynamic adaptive competitive modeling to enhance salient object detection. Its core concept, “competitive fusion,” draws inspiration from the spatial synergistic suppression–enhancement mechanism in visual cognition [18]. Research [19] indicates that the cerebral cortex achieves fine-tuned attention regulation through competitive mechanisms across different hierarchical scales, where suppressing distracting objects not only weakens their representation but also enhances responses to salient features—reflecting a dynamic process of “distractor suppression and target enhancement.”

MC2FNet emulates this cognitive principle by employing a layer-wise feature extraction architecture to progressively extract multi-scale feature representations from RGB and depth/thermal images. It then utilizes a cross-modal feature fusion update module (CMFU) to achieve efficient competitive interaction between features from different modalities. This module incorporates a spatial attention mechanism combining channel reduction (CR) and group aggregation token re-embedding (GATR), effectively integrating intra-modal, cross-modal, and multi-scale features with low computational complexity. As illustrated in Fig. 2, GATR enables multi-level fine-grained perception through grouped aggregation using different convolution kernels and implements attention re-embedding via a block-by-pix approach. These two mechanisms map features to different spatial dimensions to capture modality differences, enabling robust spatial feature extraction under adaptive competition while reducing disruptive noise. The hybrid attention strategy enhances cross-modal complementary capabilities, especially in scenarios with depth/thermal consistency or color-texture similarity, simulating neural enhancement and suppression during visual recognition. Multi-scale competitive fusion further supplements semantics across scales, promoting competitive fine-tuning of hierarchical features. By constructing a multi-scale cross-modal suppression–enhancement dynamic game, the model mimics the visual system’s coarse-to-fine feature optimization pathway. This competitive attention fusion achieves a dynamic balance between suppression and enhancement during feature representation, thereby improving selective integration and semantic consistency of multimodal information. Experimental results demonstrate that MC2FNet achieves state-of-the-art performance in salient object detection tasks.

Our main contributions are as follows:

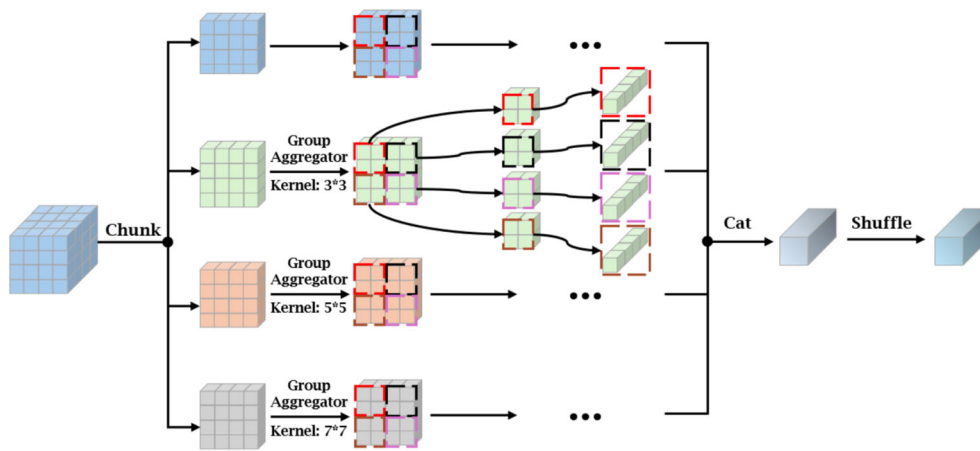


Fig. 2 Group aggregation token re-embedding (GATR). The Query, Key, and Value obtained from the linear mapping of the attention are re-embedded using GATR to accomplish multi-level fine-grained feature

aggregation perception by grouping computations with different convolutional kernels, and merging and converting traditional pixel tokens into patch tokens

- We propose a novel multi-scale cross-modal competitive fusion network (MC2FNet) for multimodal salient object detection, which emulates the spatial synergistic suppression–enhancement mechanism in visual cognition to achieve effective competitive fusion of multi-scale cross-modal features.
- Based on the spatial synergistic suppression–enhancement mechanism, we propose a competitively fused multimodal SOD decoder that simulates the feature optimization path of the visual system from coarse to fine. Based on the proposed spatial hybrid attention, this decoder accomplishes competitive fusion of intra-modal, cross-modal, and multi-scale long-range features, thus improving the selective integration and semantic consistency of multimodal information. This spatial hybrid attention achieves strong spatial feature extraction with low computational complexity.
- We conducted extensive experiments on RGB-D/T salient object detection datasets, and the results demonstrate that our method achieves state-of-the-art performance.

2 Related works

2.1 RGB-D salient object detection

Early RGB-D SOD methods relied on hand-crafted feature extraction within a traditional scale space, encompassing color, texture, edges, and contrast. The extracted features were then analyzed using machine learning algorithms for object detection. Although some algorithms achieved satisfactory performance in standard environments [20–22], their limited ability to capture deep semantic features restricted their applicability in complex scenarios, leading to reduced

accuracy in SOD and segmentation tasks. The advent of convolutional neural networks (CNNs) marked a substantial advancement in SOD development. CNN-based pixel-wise feature extraction has reformulated object detection as a binary classification problem. The predominant approach involves extracting RGB and depth features via multi-layer convolutional networks, which are then integrated using concatenation or attention mechanisms [23, 24]. However, relying solely on unimodal feature extraction poses challenges in balancing fine-grained details with high-level semantic information.

Consequently, extensive research has focused on cross-modal feature fusion in SOD. Cross-modal feature fusion strategies can be categorized according to the number of feature encoders as single-stream, dual-stream, and triple-stream architectures [25–30]. Single-stream architectures can be further classified into depth-input and depth-supervision approaches. In particular, Zhao et al. [31] proposed a single-stream architecture that directly integrates depth maps for early and mid-stage fusion between RGB and depth modalities. Wei et al. [32] adopted a depth-supervision approach that employs collaborative learning to enhance SOD. Zhang et al. [11] designed an early fusion network, FasterSal, with a single-stream architecture, maintaining high feature encoding efficiency. Despite its advantages of lower parameter complexity and reduced computational cost, the single-stream structure is limited by insufficient cross-modal fusion caused by modality inconsistency. Dual-stream architectures effectively address the challenge of cross-modal inconsistency and insufficient fusion by employing separate encoder–decoder networks [33, 34] for RGB and depth features. This approach enables complementary interactions between modalities, allowing different streams to contribute unique and complementary features. Some stud-

ies have focused on enhancing feature fusion at the encoder stage, while others integrate cross-modal assistance at the decoding stage [35–37]. Given the inherent disparities and complementary nature of different modalities, recent studies have explored shared feature learning and bidirectional interactions. For instance, Gao et al. [38] proposed a dual-stream framework that employs MobileViT as the network encoder and integrates an Adaptive Edge-aware Fusion Module (AEFM) to optimize feature fusion. Zhu et al. [39] introduced the Cross-Modal Inverse Guidance Network (CMIGNet), incorporating a multi-level function-guided enhancement mechanism to mitigate modality dependency during dual-stream interactions. Triple-stream architectures incorporate three separate encoders to extract features from RGB, depth, and RGB-D images simultaneously, facilitating a more comprehensive fusion of features [40, 41]. Zuo et al. [42] proposed a triple-stream architecture that supervises the RGB, depth, and fused modalities to generate saliency maps, effectively addressing cross-modal inconsistency. However, these existing approaches still exhibit limitations in fully capturing the intricate dependencies between RGB and depth modalities. How to design a more efficient adaptive fusion mechanism to dynamically balance semantic consistency and pattern specificity has become an urgent problem to be solved. Our answer is to draw on neural augmentation and neural suppression in visuospatial structures to fuse multimodal features by dynamically competing to enhance selective integration and semantic balance of information.

2.2 RGB-T salient object detection

Traditional RGB-T SOD methods are based primarily on machine learning techniques that use hand-crafted features for saliency detection. These approaches are heavily dependent on prior knowledge and designer expertise, often resulting in suboptimal performance in complex scenarios [43, 44]. With advancements in deep learning for computer vision, convolutional neural networks (CNNs) have become widely adopted for RGB-T SOD due to their relatively simple architecture and fewer parameters [45]. The integration of attention mechanisms has further enhanced semantic representation and fine-grained feature extraction, mitigating the limitations of CNN [46]. For example, Zhang et al. [47] used a VGG-16 backbone to extract coarse features from RGB and thermal images. They introduced the Adjacent Depth Feature Combination (ADFC) module, which captures fine details at each stage and facilitates bidirectional information propagation through attention guidance, significantly improving prediction accuracy. To further enhance feature representation, various feature refinement techniques have been proposed. Chen et al. [48] developed a Feature Difference Reduction Module, which enhances modal feature consistency while capturing discriminative information.

Although CNN-based approaches have shown promising results in SOD, they still suffer from limited global semantic awareness due to their inherent focus on local regions.

The emergence of Vision Transformers (ViT) has effectively addressed the locality constraints of CNNs, leading to superior performance in SOD. Unlike CNNs, ViTs can capture long-range dependencies, making them well suited for multimodal saliency detection. For instance, Luo et al. [49] designed a Transformer-based dual-stream encoder to extract multimodal features. In the decoder, they introduced the Modal Complementary Information Module and the Multi-Scale Adaptive Feature Fusion Module, which effectively fuse complementary modal information, locate salient regions, and suppress redundant features. Additionally, they proposed a Salient Region Mining Module (SRMM) to refine boundary features. To enhance saliency detection under low-light conditions, Yue et al. [50] developed SFMNet, a model that leverages frequency-domain features for saliency detection. They also introduced a low-light RGB-T SOD dataset to address the lack of suitable benchmark datasets. In addressing modal saliency cues, Jin et al. [51] proposed the Cross-Modal Asymmetric Feature Complementation Network (CAFCNet), which enhances and fuses RGB and thermal features, effectively reducing interference from the thermal modality. Furthermore, Tang et al. [52] introduced CONTRINET, a three-stream structure that employs a divide-and-conquer strategy to process modality-specific and complementary information separately, improving saliency detection accuracy. However, existing methods still have shortcomings in enhancing the complementary feature extraction capabilities between RGB and thermal modalities. The core challenge lies in how to design an efficient attention mechanism to achieve the extraction of long-range cross-modal semantic complementary features. Our approach utilizes channel reduction with group aggregation token re-embedding to map features to feature spaces of different dimensions to capture modal differences and achieve strong spatial long-range complementary feature extraction.

3 Methodology

In this paper, an efficient Transformer-based multimodal salient object detection network is proposed. Figure 3 shows the overall architecture of the proposed MC2FNet model. Its main components include a two-stream pre-trained ResNet [53] encoder and a multi-scale cross-modal competitive fusion Transformer decoder.

In the decoding part, we propose a Cross-Modal Fusion Update Unit (CMFU) network model that accomplishes competitive fusion within the same scale modality, between modalities, and between different scales of decoding. It

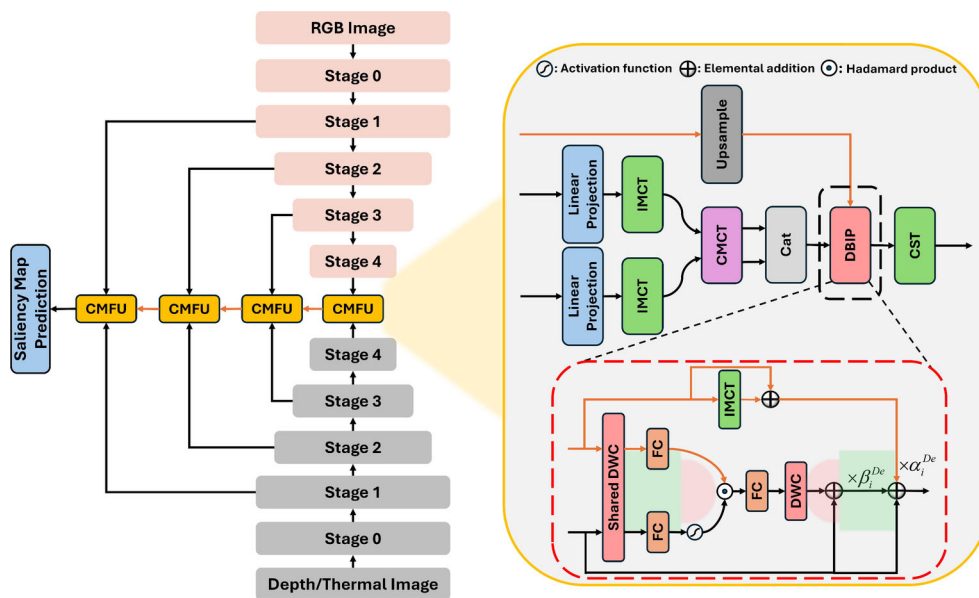


Fig. 3 Overview of the proposed MC2FNet model. It consists of a two-stream pre-trained encoder and a multi-scale cross-modal competitive fusion Transformer decoder

mainly contains the Intra-Modal Competition Transformer Module (IMCT), the Cross-Modal Competition Transformer Module (CMCT), and the Competition Module between different scales (DBIP). Finally, feature fusion is accomplished using the Cross-Scale Transformer (CST), which is a cross-scale IMCT.

3.1 Intra-modal competition transformer

While channel attention is efficient, it often overlooks the usefulness of spatial features. To address this, our proposed IMCT employs a hybrid multi-head attention mechanism to capture long-range dependencies within modalities from spatial dimensions, with its specific structure illustrated in Fig. 4. This approach utilizes two complementary attention mechanisms to extract effective features through spatial information competition.

In the IMCT module, the input bimodal features are first normalized. The spatial features within each modality are then processed via competitive learning using Mix Multi-Head Self-Attention (Mix-MHSA), which integrates Channel Reduction Attention (CRA) and Group Aggregation Re-Embedding Spatial Attention (GARSA), to accomplish spatial long-range feature extraction. Before the similarity computation in the multi-head attention-based CRA, we incorporate implicit condition-based positional encoding (PEG) [54], enhancing its flexibility in handling input sequences of varying lengths. Features endowed with this implicit conditional positional encoding are further processed through average pooling and pointwise convolution to extract salient information, yielding the initial key and

value. The final unidimensional query and key are derived via linear dimensionality reduction mapping, while the value is obtained through equi-dimensional linear mapping. This design reduces computational cost in subsequent attention operations and allows CRA to focus more on global spatial information. It is summarized in Equation (1).

$$\begin{cases} f_i^j = f_i^j + PEG(f_i^j) \\ Q_C^{ij} = f_i^j W_{Cq}^{ijh} \\ K_C^{ij} = Conv_{1 \times 1}(AvgPool(f_i^j)) W_{Ck}^{ijh} \\ V_C^{ij} = Conv_{1 \times 1}(AvgPool(f_i^j)) W_{Cv}^{ijh} \\ Attn(Q_C^{ij}, K_C^{ij}, V_C^{ij}) = \sigma \left(\frac{Q_C^{ij} K_C^{ijT}}{\sqrt{d}} \right) V_C^{ij} \\ CRA(f_i^j) = [Attn_h(Q_C^{ij}, K_C^{ij}, V_C^{ij})]_{h=1}^{Nh} W_o^{ij} \end{cases} \quad (1)$$

where f_i^j are the different modal features of the input, $i \in \{CMFU_1, CMFU_2, CMFU_3, CMFU_4\}$, and $j \in \{rgb, depth/thermal\}$. $W_{Cq}^{ij} \in \mathbb{R}^{C_i \times 1}$, $W_{Ck}^{ij} \in \mathbb{R}^{C_i \times \frac{C_i}{h}}$, and $W_o^{ij} \in \mathbb{R}^{C_i \times C_i}$ are linear projection parameters, σ represents the Softmax function, and $[...]_{h=1}^{Nh}$ stands for the head-connection operation. $AvgPool$ is the average pooling of scale p_i at each stage, CRA is an efficient attention compared to traditional attention, and its computational complexity is as follows:

$$\begin{cases} N^p = \frac{N}{p_i} \\ \Omega(CRA) = (N^p)^2 1 + (N^p)^2 C \end{cases} \quad (2)$$

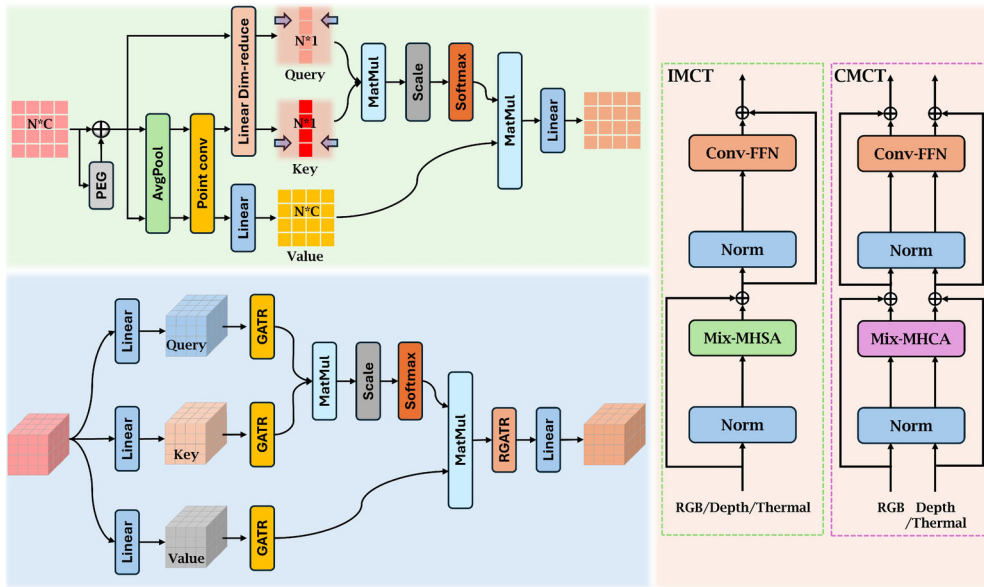


Fig. 4 Our proposed Intra-Modal Competitive Transformer (IMCT) and Cross-Modal Competitive Convergent Transformer (CMCT) models are shown on the right. The top left is the channel reduction attention (CRA), and the bottom left is the group aggregation re-embedding

spatial attention (GARSA), which utilize the competing factor compositions Mix-MHSA and Mix-MHCA to achieve intra-modal and cross-modal feature fusion, respectively

where N denotes the number of pixel tokens, and the left and right terms represent the computation of the query-key similarity score and the attention weights, respectively. Additionally, we design a spatial attention mechanism called GARSA, which utilizes group aggregation token re-embedding (GATR). This mechanism employs convolutional kernels of different sizes to achieve fine-grained attention across different channel groups. The tokens obtained from each group are spliced via token re-embedding operations, and group weight interaction is facilitated through shuffling, as shown in Fig. 2. The batch aggregation process of GATR is shown in Equation (3). Taking input feature x as an example, we divide the feature into four groups along the channel dimension and process them using chunking. Group-wise feature extraction is performed using convolutional kernels of different sizes and pointwise convolutions. Batch normalization (BN) is applied to each group.

$$\begin{cases} x_1, x_2, x_3, x_4 = \text{Chunk}(x) \\ x'_2 = \text{BN}(\text{Conv}_{1 \times 1}(\text{Conv}_{3 \times 3}(x_2))) \\ x'_3 = \text{BN}(\text{Conv}_{1 \times 1}(\text{Conv}_{5 \times 5}(x_3))) \\ x'_4 = \text{BN}(\text{Conv}_{1 \times 1}(\text{Conv}_{7 \times 7}(x_4))) \end{cases} \quad (3)$$

This is followed by a block-by-pix re-embedding operation. At this stage, the features in each group can be represented as: $\mathbb{R}^{(Nh \cdot hd) \times (Nhp \cdot ph) \times (Nwp \cdot pw)} \rightarrow \mathbb{R}^{Nh \times (hd \cdot ph \cdot pw) \times (Nhp \cdot Nwp)}$, where Nh denotes the number of attention heads, hd is the dimension of each attention head, Nhp is the number of patches in the height direction, ph is the patch height, nwp

is the number of patches in the width direction, and pw is the patch width. The aggregated features after group mapping are concatenated, and information exchange between different groups is achieved through Shuffle. As shown in Equation (4), $[\dots, \dots]$ represents the concatenation operation, and x'_1, x'_2, x'_3, x'_4 denote the grouped features after re-embedding mapping. Here, x' represents the output features of GATR. In practice, the query, key, and value for the attention mechanism are derived from this output.

$$x' = \text{Shuffle}[x'_1, x'_2, x'_3, x'_4] \quad (4)$$

Subsequently, the query, key, and value embeddings are constructed to complete the spatial attention computation. Finally, a reverse GATR (RGATR) operation is applied to the resulting tokens to produce the final output. The entire GARSA calculation process is summarized in Equation (5).

$$\begin{cases} Q_G^{ij} = \text{GATR}(f_i^j W_{Gq}^{ijh}) \\ K_G^{ij} = \text{GATR}(f_i^j W_{Gk}^{ijh}) \\ V_G^{ij} = \text{GATR}(f_i^j W_{Gv}^{ijh}) \\ \text{Attn}(Q_G^{ij}, K_G^{ij}, V_G^{ij}) = \sigma \left(\frac{Q_G^{ij} K_G^{ijT}}{\sqrt{d}} \right) V_G^{ij} \\ \text{GARSA}(f_i^j) = \text{RGATR}([\text{Attn}_h(Q_G^{ij}, K_G^{ij}, V_G^{ij})]_{h=1}^{Nh}) W_{Go}^{ij} \end{cases} \quad (5)$$

where $W_{Gq}^{ijh}, W_{Gk}^{ijh}, W_{Gv}^{ijh} \in \mathbb{R}^{C_i \times C_i}$ are the corresponding projection matrices. $\sqrt{d}$ is the scale factor. GARSA uses

the GATR method to reduce computational complexity, thus ensuring that the model is able to lightly integrate a wider range of contextual information. GARSA eschews the traditional pixel-by-pixel computation of matrix operations in favor of a block-by-pix form of computation with a wider receptive field, thus reducing complexity by p^2 , which is the number of elements in the patch. We assign corresponding competing factors to CRA and GARSA, and Mix-MHSA can then be expressed as:

$$MixMHSA(f_i^j) = \alpha_s^{ij} CRA(f_i^j) + \beta_s^{ij} GARSA(f_i^j) \quad (6)$$

Competition factors α_s^{ij} and β_s^{ij} are learnable parameters constrained within the interval (0, 1) via the Sigmoid activation function. Initially set to 0.5, they undergo adaptive updates through backpropagation to achieve dynamic fusion within the modality. Therefore, the IMCT can be expressed by Equation (7).

$$\begin{cases} f_i^j = MixMHSA(Norm(f_i^j)) + f_i^j \\ IMCT(f_i^j) = Conv_FFN(Norm(f_i^j)) + f_i^j \end{cases} \quad (7)$$

where $Conv_FFN$ is the use of convolutional operations to adjust the feed-forward network (FFN) to compensate for the lack of local attention.

3.2 Cross-modal competition transformer

Figure 4 illustrates the proposed Cross-Modal Competitive Fusion Transformer (CMCT), a fusion module based on cross-modal competitive attention. This module utilizes the two previously introduced attention mechanisms to achieve cross-competition between features of different modalities, enabling long-range attention and effective fusion of cross-modal features. The cross-attention formulations for CRA and GARSCA across different modalities are given as follows:

$$\begin{cases} CRA_r(Q_C^{ir}, K_C^{id/t}, V_C^{id/t}) = \sigma\left(\frac{Q_C^{ir} K_C^{id/tT}}{\sqrt{d}}\right) V_C^{id/t} \\ CRA_{d/t}(Q_C^{id/t}, K_C^{ir}, V_C^{ir}) = \sigma\left(\frac{Q_C^{id/t} K_C^{irT}}{\sqrt{d}}\right) V_C^{ir} \\ GARSCA_r(Q_G^{ir}, K_G^{id/t}, V_G^{id/t}) = \sigma\left(\frac{Q_G^{ir} K_G^{id/tT}}{\sqrt{d}}\right) V_G^{id/t} \\ GARSCA_{d/t}(Q_G^{id/t}, K_G^{ir}, V_G^{ir}) = \sigma\left(\frac{Q_G^{id/t} K_G^{irT}}{\sqrt{d}}\right) V_G^{ir} \end{cases} \quad (8)$$

Then, the attention of hybrid cross-modal cross-competition can be expressed by Equation (9).

$$\begin{cases} MixMHCA_r = \alpha_c^{ir} CRA_r(Q_C^{ir}, K_C^{id/t}, V_C^{id/t}) \\ \quad + \beta_c^{ir} GARSCA_r(Q_G^{ir}, K_G^{id/t}, V_G^{id/t}) \\ MixMHCA_{d/t} = \alpha_c^{id/t} CRA_{d/t}(Q_C^{id/t}, K_C^{ir}, V_C^{ir}) \\ \quad + \beta_c^{id/t} GARSCA_{d/t}(Q_G^{id/t}, K_G^{ir}, V_G^{ir}) \end{cases} \quad (9)$$

where $MixMHCA_r$ denotes the result of cross-modal competing attention for RGB modality and $MixMHCA_{d/t}$ denotes the result of cross-modal competing attention for depth/thermal modality. The initialization and update methods for competition factors in cross-modal fusion mirror those applied to intra-modal competition factors. The other subsequent calculation steps follow the same process as the IMCT calculation.

3.3 Different scales competition fusion module

Following cross-modal competitive fusion, multi-scale feature fusion is carried out via the DBIP module, whose structure is depicted in Fig. 3. Larger-scale features capture the overall shape and contour of the target, while smaller-scale features provide finer local details. It is therefore essential to integrate features across scales to extract more comprehensive semantic information. The lower-scale features are up-sampled for scale alignment. These aligned multi-scale features then undergo shared depth-wise convolution (DWC), which enables parameter sharing to enhance learning efficiency and performance while reducing parameter count. Fully connected (FC) mapping is employed to capture richer representations in high-dimensional space, and cross-scale weight assignment is performed via the Hadamard product. The resulting weighted cross-scale features are fused with the filtered low-scale features to complete the final competitive fusion. The computational procedure of DBIP is presented in Equation (10).

$$\begin{cases} f_i^{f1} = ReLU(FC(DWC_s(f_i^{fc}))) \odot FC(DWC_s(f_{i-1}^f)) \\ f_i^{f2} = f_i^{fc} + DWC(FC(f_i^{f1})) \\ f_i^f = f_i^{fc} + \alpha_i^{De}(IMCT(f_{i-1}^f) + f_{i-1}^f) + \beta_i^{De} f_i^{f2} \end{cases} \quad (10)$$

Among these, f_i^{fc} represents the connection features after cross-modal fusion via CMCT, while f_i^{f1} and f_i^{f2} denote intermediate results during fusion. f_{i-1}^f denotes the fusion output feature from the previous layer, f_i^f denotes the current-layer feature fused with cross-scale information, and DWC_s denotes shared deep convolution. α_i^{De} and β_i^{De}

denote competition factors, introduced to compensate for the lack of local granularity in attention. Similarly, the update mechanisms for competition factors α_i^{De} and β_i^{De} remain consistent with those employed in IMCT and CMCT. Specifically, DBIP employs a dual-branch interactive architecture to enable adaptive multimodal feature fusion and enhancement across different scales. This design is built upon a shared dual-stream DWC architecture, thereby enabling efficient bidirectional, multi-scale feature interaction and information complementarity between different modalities. This framework facilitates the effective integration and representation of multimodal cues at various scales, thereby enhancing the network's ability to distinguish features in SOD.

4 Experimental results and analysis

4.1 Experimental setup

4.1.1 Dataset and baseline

We validate the proposed MC2FNet model on three publicly available RGB-T and six RGB-D datasets. The three RGB-T datasets are VT821 [43] (821 image pairs), VT1000 [55] (1,000 image pairs), and VT5000 [56] (5,000 image pairs). The six RGB-D datasets include NJUD [57] (1,985 image pairs), NLPR [58] (1,000 image pairs), LFSD [59] (100 image pairs), DUTLF-Depth [60] (1,200 image pairs), STERE [61] (1,000 image pairs), and SIP [62] (929 image pairs). Since MC2FNet is designed based on a CNN backbone network, we also perform a comprehensive evaluation against state-of-the-art (SOTA) methods using CNN backbones. For RGB-T SOD, we compare with fourteen methods, including ConTriNet [52], SACNet [12], and CAVER [63]. For RGB-D SOD, we include fourteen methods such as FasterSal [11], HiDANet [64], and CIRNet [29]. Additionally, three traditional RGB-T SOD methods are compared: MTMR [43], M3S-NIR [65], and SGDL [55].

4.1.2 Evaluation metrics

We evaluated the MC2FNet model using four widely used evaluation metrics. Maximum F-measure ($maxF$) [80] evaluates the accuracy and recall under the optimal threshold, Maximum enhanced-alignment measure E_{max}^{ξ} [81] simultaneously considers pixel-level errors and image-level errors, the structure measure S_m [82] evaluates region and object-aware structural similarity, and the mean absolute error (MAE) [80] calculates the average absolute error per pixel.

4.1.3 Implementation details

During training, we employed the SGD optimizer with an initial learning rate of 0.005 and a weight decay coefficient of 0.0005 to update model parameters. During MC2FNet training, data augmentation was achieved through strategies such as spatial transformations, horizontal flipping, and color dithering. All experiments were conducted on an NVIDIA RTX 2080 Ti GPU with a batch size of 8, completing 100 training epochs. To ensure a fair comparison, the RGB-T SOD training set consists of only 2,500 samples from VT5000TR, with all remaining data used for testing. For RGB-D SOD, we employed two common training configurations. In the first configuration, we used 1,485 samples from NJUD and 700 samples from NLPR for training, and the rest as the test set. Test results are shown as MC2FNet (I) in Table 2. In the second configuration, we used 1,485 samples from NJUD, 700 from NLPR, and 800 from DUTLF-Depth for training, with the remaining data reserved for testing. Test results are shown as MC2FNet (II) in Table 2.

4.2 Analysis of results

4.2.1 Quantitative analysis

As shown in Table 1 and Table 2, MC2FNet demonstrates outstanding performance. Table 1 presents a comparison of MC2FNet with 17 state-of-the-art (SOTA) RGB-T SOD methods across three datasets, while Table 2 evaluates MC2FNet against 14 SOTA RGB-D SOD methods across six datasets. The top-ranked results are highlighted in Bold, while the second-ranked ones are highlighted in *Italic*. Higher values of S_m , $maxF$, and E_{max}^{ξ} indicate better performance, whereas for MAE , lower values are preferable. In Table 1, MC2FNet achieves top performance in 10 out of 12 metrics across the three datasets, ranking second in one metric and third in another. MC2FNet ranks third on the VT821 dataset in terms of the E_{max}^{ξ} metric, only behind MGAI and ConTriNet50. MGAI effectively reduces both pixel-level and image-level errors by fusing dual-modal features with edge features, whereas our model currently lacks detailed guidance from boundary features. On the VT5000 dataset, MC2FNet achieves the second-best performance in MAE , trailing only ConTriNet50, which employs a divide-and-conquer strategy. This approach enables in-depth exploration of modality-specific cues while achieving effective fusion of complementary information across modalities, thereby improving detection performance in challenging scenes. Similarly, in Table 2, MC2FNet attains first-

Table 1 Comparison of detection performance with seventeen state-of-the-art models on three RGB-T datasets, containing fourteen CNN backbone network-based methods, and three traditional methods

Methods	VT821 [43]					VT1000 [55]					VT5000 [56]				
	$S_m \uparrow$	$MAE \downarrow$	$max F \uparrow$	$E_{max}^k \uparrow$		$S_m \uparrow$	$MAE \downarrow$	$max F \uparrow$	$E_{max}^k \uparrow$		$S_m \uparrow$	$MAE \downarrow$	$max F \uparrow$	$E_{max}^k \uparrow$	
MTMR [43]	0.7252	0.1084	0.6897	0.8117		0.7057	0.1194	0.7154	0.8359		0.6801	0.1143	0.6133	0.7923	
M3S-NIR [65]	0.7231	0.1397	0.7378	0.8370		0.7258	0.1454	0.7354	0.8279		0.6521	0.1680	0.5957	0.7599	
SGDL [55]	0.7654	0.0849	0.7348	0.8395		0.7867	0.0896	0.7700	0.8585		0.7504	0.0886	0.6949	0.8292	
ADF [56]	0.8103	0.0765	0.7522	0.8390		0.9095	0.0339	0.9081	0.9505		0.8635	0.0482	0.8369	0.9111	
MIDD [66]	0.8711	0.0445	0.8508	0.9183		0.9067	0.0293	0.9058	0.9523		0.8557	0.0460	0.8384	0.9131	
CSRNet [67]	0.8848	0.0375	0.8579	0.9227		0.9184	0.0242	0.9083	0.9526		0.8676	0.0417	0.8370	0.9138	
CGFNet [46]	0.8805	0.0378	0.8658	0.9205		0.9233	0.0231	0.9235	0.9594		0.8830	0.0353	0.8690	0.9265	
MMNet [68]	0.8733	0.0397	0.8480	0.9206		0.9139	0.0269	0.9064	0.9526		0.8621	0.0434	0.8364	0.9141	
MIA-DPD [69]	0.8444	0.0699	0.8306	0.9018		0.9237	0.0251	0.9279	0.9645		0.8785	0.0404	0.8651	0.9295	
ECFNet [70]	0.8766	0.0345	0.8346	0.9104		0.9232	0.0215	0.9168	0.9592		0.8741	0.0377	0.8475	0.9213	
MGAI [71]	0.8909	0.0311	0.8726	0.9354		0.9289	0.0212	0.9256	0.9662		0.8833	0.0345	0.8617	0.9314	
Scribble [72]	0.8571	0.0359	0.8321	0.9081		0.9056	0.0266	0.9038	0.9486		0.8434	0.0423	0.8146	0.9040	
LSNet [73]	0.8787	0.0332	0.8448	0.9206		0.9257	0.0227	0.9216	0.9626		0.8773	0.0371	0.8497	0.9240	
CAVER [63]	0.8901	0.0384	0.8738	0.9286		0.9376	0.0180	0.9368	0.9719		0.8965	0.0324	0.8767	0.9367	
SACNet [12]	0.8829	0.0325	0.8543	0.9203		0.9325	0.0177	0.9279	0.9657		0.8919	0.0302	0.8703	0.9344	
ConTriNet16 [52]	0.8811	0.0309	0.8470	0.9182		0.9243	0.0199	0.9183	0.9636		0.8828	0.0325	0.8630	0.9283	
ConTriNet50 [52]	0.8924	0.0274	0.8663	0.9318		0.9292	0.0189	0.9257	0.9682		0.8935	0.0298	0.8781	0.9384	
MC2FNet	0.8952	0.0268	0.8746	0.9301		0.9422	0.0153	0.9435	0.9753		0.9014	0.0302	0.8832	0.9414	

The top-ranked results are highlighted in Bold, while the second-ranked ones are highlighted in Italic

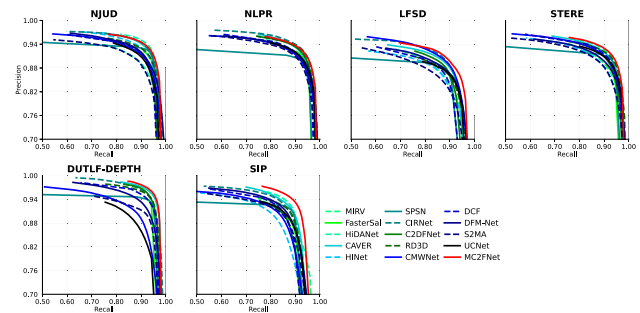
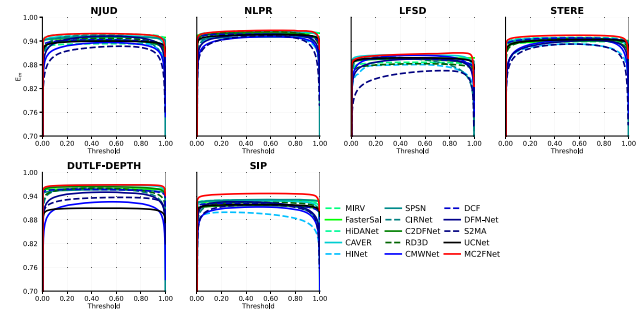
Table 2 Detection performance comparison with fourteen state-of-the-art models based on CNN backbone networks on six RGB-D datasets

Datasets	CNN backbone networks														Ours		
	CVPR20	CVPR20	ECCV20	CVPR21	AAAI21	MM21	TMM22	ECCV22	TIP22	PR23	TIP23	CAVER	HIDANet	FasterSal	MIRV	MC2FNet	
	S2MA [74]	UCNet [26]	CMWNet [27]	DCF [75]	RD3D [76]	DFM-Net [77]	C2DFNet [28]	SPSN [78]	CIRNet [29]	HiNet [79]	[63]	[63]	[64]	[11]	[30]	(I)	(II)
NJUD [57]	$S_m \uparrow$	0.8867	0.8974	0.9029	0.9037	0.9156	0.9072	0.9079	0.9191	0.9250	0.9150	0.9227	0.9258	0.9100	0.8896	0.9240	0.9270
	$MAE \downarrow$	0.0577	0.0413	0.0456	0.0387	0.0365	0.0428	0.0389	0.0330	0.0350	0.0386	0.0315	0.0292	0.0344	0.0456	0.0304	0.0279
	$maxF \uparrow$	0.8816	0.8962	0.9025	0.9052	0.9139	0.9130	0.9086	0.9193	0.9277	0.9135	0.9240	0.9314	0.9126	0.8891	0.9277	0.9278
	$E_{max}^{\xi} \uparrow$	0.9268	0.9397	0.9362	0.9431	0.9471	0.9525	0.9423	0.9505	0.9546	0.9445	0.9535	0.9568	0.9498	0.9333	0.9581	0.9588
	$S_m \uparrow$	0.9155	0.9195	0.9171	0.9220	0.9296	0.9230	0.9279	0.9171	0.9334	0.9222	0.9276	0.9294	0.9221	0.9138	0.9292	0.9326
NLPR [58]	$MAE \downarrow$	0.0298	0.0250	0.0293	0.0235	0.0221	0.0261	0.0217	0.0263	0.0227	0.0257	0.0236	0.0212	0.0226	0.0253	0.0250	0.0214
	$maxF \uparrow$	0.9017	0.9034	0.9030	0.9099	0.9190	0.9078	0.9166	0.8981	0.9241	0.9062	0.9197	0.9232	0.9110	0.9042	0.9233	0.9224
	$E_{max}^{\xi} \uparrow$	0.9532	0.9560	0.9510	0.9574	0.9649	0.9572	0.9605	0.9507	0.9656	0.9565	0.9637	0.9633	0.9605	0.9569	0.9620	0.9669
	$S_m \uparrow$	0.8292	0.8564	0.8680	0.8481	0.8500	0.8574	0.8557	0.8505	0.8673	0.8445	0.8667	–	0.8538	0.8409	0.8550	0.8749
	$MAE \downarrow$	0.1018	0.0738	0.0737	0.0783	0.0806	0.0792	0.0727	0.0736	0.0750	0.0834	0.0690	–	0.0707	0.0796	0.0804	0.0671
LFSD [59]	$maxF \uparrow$	0.8310	0.8595	0.8784	0.8559	0.8501	0.8598	0.8629	0.8629	0.8788	0.8428	0.8760	–	0.8545	0.8505	0.8625	0.8864
	$E_{max}^{\xi} \uparrow$	0.8652	0.8975	0.9040	0.8957	0.8828	0.8955	0.8947	0.8966	0.9009	0.8807	0.9049	–	0.8974	0.8863	0.8948	0.9100
	$S_m \uparrow$	0.8904	0.9029	0.9050	0.9057	0.9113	0.8980	0.9023	0.9067	–	0.8918	0.9096	0.9114	0.8903	0.8906	0.9148	0.9173
	$MAE \downarrow$	0.0507	0.0391	0.0433	0.0370	0.0374	0.0446	0.0385	0.0372	–	0.0490	0.0365	0.0353	0.0405	0.0416	0.0341	0.0323
	$maxF \uparrow$	0.8823	0.8990	0.9009	0.9044	0.9061	0.8925	0.8973	0.9006	–	0.8828	0.9073	0.9117	0.8840	0.8880	0.9129	0.9160
DUTLF-Depth [60]	$E_{max}^{\xi} \uparrow$	0.9324	0.9438	0.9444	0.9477	0.9466	0.9413	0.9430	0.9450	–	0.9325	0.9483	0.9492	0.9411	0.9387	0.9530	0.9546
	$S_m \uparrow$	0.9034	0.8712	0.8866	0.9241	0.9316	0.9129	0.9326	0.9256	0.9323	–	0.9389	–	0.9185	–	0.9040	0.9428
	$MAE \downarrow$	0.0435	0.0586	0.0562	0.0303	0.0306	0.0385	0.0252	0.0302	0.0309	–	0.0252	–	0.0300	–	0.0454	0.0233
	$maxF \uparrow$	0.9005	0.8661	0.8883	0.9318	0.9389	0.9179	0.9440	0.9294	0.9402	–	0.9499	–	0.9294	–	0.9146	0.9523
	$E_{max}^{\xi} \uparrow$	0.9368	0.9096	0.9256	0.9571	0.9598	0.9500	0.9638	0.9563	0.9605	–	0.9675	–	0.9578	–	0.9391	0.9684
SIP [62]	$S_m \uparrow$	0.8721	0.8751	0.8674	0.8735	0.8853	0.8831	0.8715	0.8878	0.8884	0.8561	0.8906	0.8921	0.8725	0.8764	0.8963	0.9068
	$MAE \downarrow$	0.0569	0.0514	0.0620	0.0519	0.0484	0.0507	0.0529	0.0454	0.0523	0.0656	0.0446	0.0438	0.0493	0.0491	0.0406	0.0359
	$maxF \uparrow$	0.8771	0.8792	0.8741	0.8859	0.8890	0.8873	0.8770	0.8959	0.8959	0.8550	0.9001	0.9040	0.8783	0.8788	0.9101	0.9200
	$E_{max}^{\xi} \uparrow$	0.9185	0.9186	0.9126	0.9218	0.9243	0.9259	0.9160	0.9314	0.9240	0.8993	0.9290	0.9296	0.9295	0.9283	0.9393	0.9467

The top-ranked results are highlighted in Bold, while the second-ranked ones are highlighted in Italic

Table 3 To assess MC2FNet's stability, we performed five repeated experiments on the RGB-D SOD task and reported the average values for each metric

Number	LFSD				STERE				DUTLF-Depth				SIP			
	$S_m \uparrow$	$MAE \downarrow$	$maxF \uparrow$	$E_{max}^E \uparrow$	$S_m \uparrow$	$MAE \downarrow$	$maxF \uparrow$	$E_{max}^E \uparrow$	$S_m \uparrow$	$MAE \downarrow$	$maxF \uparrow$	$E_{max}^E \uparrow$	$S_m \uparrow$	$MAE \downarrow$	$maxF \uparrow$	$E_{max}^E \uparrow$
1	0.8662	0.0712	0.8703	0.9009	0.9143	0.0333	0.9132	0.9522	0.9419	0.0231	0.9550	0.9701	0.9085	0.0347	0.9232	0.9493
2	0.8709	0.0670	0.8813	0.9083	0.9142	0.0329	0.9118	0.9515	0.9424	0.0232	0.9536	0.9694	0.9050	0.0353	0.9201	0.9476
3	0.8696	0.0678	0.8746	0.9046	0.9171	0.0327	0.9157	0.9532	0.9449	0.0218	0.9555	0.9708	0.9057	0.0361	0.9210	0.9460
4	0.8777	0.0638	0.8829	0.9137	0.9170	0.0320	0.9175	0.9552	0.9428	0.0218	0.9539	0.9703	0.9082	0.0349	0.9214	0.9496
5	0.8749	0.0671	0.8864	0.9100	0.9173	0.0323	0.9160	0.9546	0.9428	0.0233	0.9523	0.9684	0.9068	0.0359	0.9200	0.9467
Avg	0.8719 $\pm$ 0.0058	0.0674 $\pm$ 0.0038	0.8791 $\pm$ 0.0088	0.9075 $\pm$ 0.0066	0.9160 $\pm$ 0.0018	0.0326 $\pm$ 0.0007	0.9148 $\pm$ 0.0030	0.9533 $\pm$ 0.0019	0.9430 $\pm$ 0.0019	0.0226 $\pm$ 0.0008	0.9541 $\pm$ 0.0018	0.9698 $\pm$ 0.0014	0.9068 $\pm$ 0.0018	0.0354 $\pm$ 0.0007	0.9211 $\pm$ 0.0021	0.9478 $\pm$ 0.0018

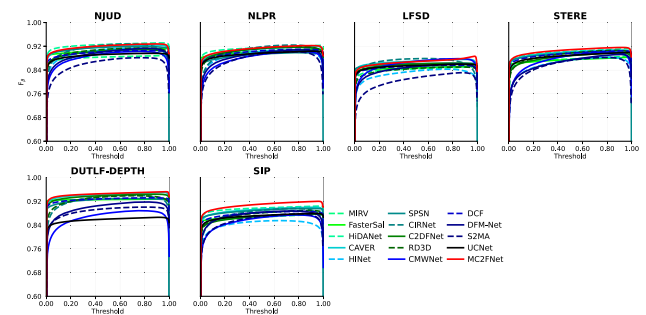
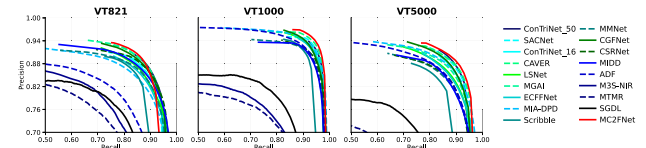
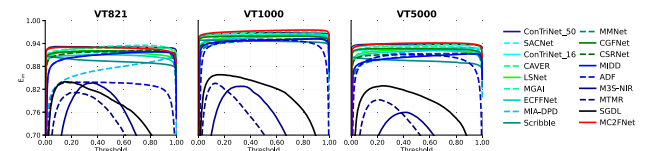
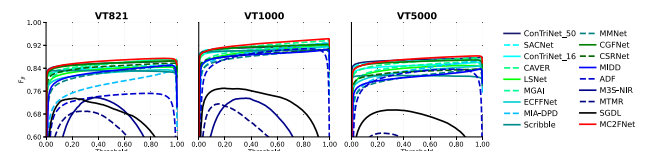
**Fig. 5** Precision–recall (PR) curves of MC2FNet against fifteen techniques on six RGB-D datasets**Fig. 6** E_m -Threshold curves of MC2FNet against fifteen techniques on six RGB-D datasets

and second-place rankings on most metrics across the six datasets. MC2FNet ranks second only to CIRNet on the NLPR dataset. CIRNet's leading performance is primarily attributed to its progressive attention-guided fusion module and importance-gated fusion module, which facilitate multi-stage cross-modal interaction at the encoder and decoder stages, respectively. MC2FNet lacks cross-modal interaction between the encoding and decoding stages. To evaluate MC2FNet's stability, we performed five repeated experiments on the RGB-D SOD task. The results in Table 3 demonstrate that the four test metrics exhibit minimal fluctuation, reflecting overall robust stability and reliability. Figures 5 and 8 illustrate the precision–recall (PR) curves for MC2FNet compared to other SOTA methods across different datasets. Higher PR curves indicate superior results, further validating MC2FNet's strong performance. Similarly, the higher E_m -Threshold and F_β -Threshold curves shown in Figs. 6, 7, 9 and 10 indicate the excellent competitiveness of our model. These results confirm MC2FNet's exceptional robustness in multimodal SOD, demonstrating its ability to accurately detect salient objects across diverse scenarios.

To further validate the superior performance of MC2FNet, we conducted a statistical significance analysis using the Wilcoxon signed-rank test to compare the performance differences between the model and the comparison methods. The specific results are shown in Table 4. To ensure the reliability and rigor of the test results, in the RGB-D modality

Table 4 Wilcoxon test between MC2FNet and twenty-seven comparison method

Methods	MTMR	M3S-NIR	SGDL	ADF	MIDD	CSRNet	CGFNet	MMNet	MIA-DPD
P value	0.0068	0.0210	0.0068	0.0342	0.0161	0.0269	0.0122	0.0161	0.0425
Significance ($p < 0.05$)	✓	✓	✓	✓	✓	✓	✓	✓	✓
Methods	ECFFNet	MGAI	Scribble	LSNet	CAVER	SACNet	ConTriNet16	ConTriNet50	S2MA
P value	0.0068	0.1099	0.0068	0.0068	0.1294	0.0076	0.0068	0.0269	0.0014
Significance ($p < 0.05$)	✓	✗	✓	✓	✗	✓	✓	✓	✓
Methods	UCNet	CMWNet	DCF	RD3D	DFM-Net	C2DFNet	SPSN	CAVER	FasterSal
P value	0.0016	0.0087	0.0006	0.0048	0.0025	0.0011	0.0031	0.0018	0.0002
Significance ($p < 0.05$)	✓	✓	✓	✓	✓	✓	✓	✓	✓

**Fig. 7** F_β -Threshold curves of MC2FNet against fifteen techniques on six RGB-D datasets**Fig. 8** Precision-recall (PR) curves of MC2FNet against seventeen techniques on three RGB-T datasets**Fig. 9** E_m -Threshold curves of MC2FNet against seventeen techniques on three RGB-T datasets**Fig. 10** F_β -Threshold curves of MC2FNet against seventeen techniques on three RGB-T datasets

experiments, four comparison methods that did not use the entire test dataset or had missing data were excluded. Only comparison models that could provide performance metrics for a complete dataset were retained for testing, thereby ensuring the completeness and validity of the sample data. Using $p < 0.05$ as the significance threshold, we tested the valid comparison methods across both modalities and found that 25 out of 27 methods had p values less than 0.05 compared to the proposed model in their respective modalities, reaching statistical significance. These results demonstrate that MC2FNet's overall performance in both modalities is significantly superior to all valid comparison methods, with its performance improvement exhibiting clear statistical significance, thereby validating the rationality and effectiveness of the proposed model design.

Table 5 Ablation study of the proposed components on the RGB-D datasets. Best results are shown in Bold

Models	NJUD			NLPR			SIP			STERE			DUTLF-Depth		
	$S_m \uparrow$	$MAE \downarrow$	$max F \uparrow$	$E_{max}^E \uparrow$	$S_m \uparrow$	$MAE \downarrow$	$max F \uparrow$	$E_{max}^E \uparrow$	$S_m \uparrow$	$MAE \downarrow$	$max F \uparrow$	$E_{max}^E \uparrow$	$S_m \uparrow$	$MAE \downarrow$	$max F \uparrow$
Ec+LP	0.7927	0.1079	0.7631	0.8500	0.8123	0.0700	0.7632	0.8722	0.7816	0.1166	0.7970	0.8795	0.7516	0.1111	0.7151
Ec+LP+IMCT	0.8958	0.0463	0.8913	0.9327	0.9129	0.0280	0.8961	0.9523	0.8541	0.0652	0.8537	0.8987	0.8669	0.0554	0.8662
Ec+LP+IMCT +CMCT	0.8984	0.0455	0.8951	0.9336	0.9151	0.0271	0.9036	0.9562	0.8607	0.0615	0.8627	0.9026	0.8762	0.0505	0.8770
Ec+LP+IMCT +CMCT+CST	0.9095	0.0399	0.9058	0.9434	0.9192	0.0271	0.9054	0.9535	0.8789	0.0551	0.8763	0.9169	0.8962	0.0424	0.8966
Ec+LP+IMCT +CMCT+CST +Add	0.9241	0.0297	0.9248	0.9550	0.9266	0.0256	0.9180	0.9625	0.9049	0.0368	0.9156	0.9449	0.9168	0.0323	0.9147
Ec+LP+IMCT +CMCT+CST +DBIP	0.9270	0.0279	0.9278	0.9588	0.9326	0.0214	0.9224	0.9669	0.9068	0.0359	0.9200	0.9467	0.9173	0.0323	0.9160
													0.9546	0.9428	0.9523
															0.9684

4.2.2 Qualitative analysis

We selected visual comparison results for indoor and outdoor environments from the RGB-T VT5000 dataset and the RGB-D NJUD dataset. As shown in Fig. 11 for RGB-T and Fig. 12 for RGB-D, MC2FNet demonstrated superior performance, with four representative image sets selected from each dataset for comparison with state-of-the-art methods. The RGB-T visual comparisons include objects affected by lighting conditions and reflections (Row 2), objects with strong background interference (Row 2, and 4), small blurry enclosed objects (Row 1), and interfering objects with similar or adjacent thermal radiation to the detected objects (Row 1-4). As shown by the VT5000 dataset results in Table 1, MC2FNet achieves the best performance across three metrics, including E_{max}^E and $max F$, while maintaining the lowest pixel-level and image-level errors, as well as the highest precision and recall rates. Furthermore, its output exhibits the S_m metric closest to the ground truth, which is visually corroborated by the clear detection of objects such as the enclosed water pipe and the bear in Fig. 11. Meanwhile, the RGB-D visual comparisons contain objects with strong interference from background color and texture similarity (Rows 2 and 3), complex detected objects (Row 4), and detected objects with depth values similar to those of interfering objects (Rows 1, 3, and 4). According to the NJUD dataset results presented in Table 2, MC2FNet achieves optimal performance across three metrics: E_{max}^E , MAE , and $max F$. As specifically visualized in Fig. 12, the model also maintains the lowest pixel-level and image-level errors, demonstrates the best structural similarity in complex scenes, and overall attains the highest precision and recall rates. These visual comparisons provide strong evidence supporting the reliability of our proposed competitive dynamic adaptive fusion mechanism, which effectively integrates complementary features from both RGB-T and RGB-D modalities and mitigates interference from distracting features.

4.3 Ablation studies

4.3.1 Effective CMFU modules

We performed ablation experiments on MC2FNet to demonstrate the effectiveness of CMFU, and as shown in Table 5, the individual components of MC2FNet contribute to achieving optimal performance. Here, Ec denotes the pre-trained encoder, and LP represents the linear projection. To evaluate the effectiveness of DBIP, we compared it with multi-scale addition in the ablation study. The ablation results indicate that the intra-modal competitive fusion approach for different spatial dimensions can effectively enhance the extraction of strong spatial features, thereby improving intra-modal semantic understanding. The cross-modal long-range

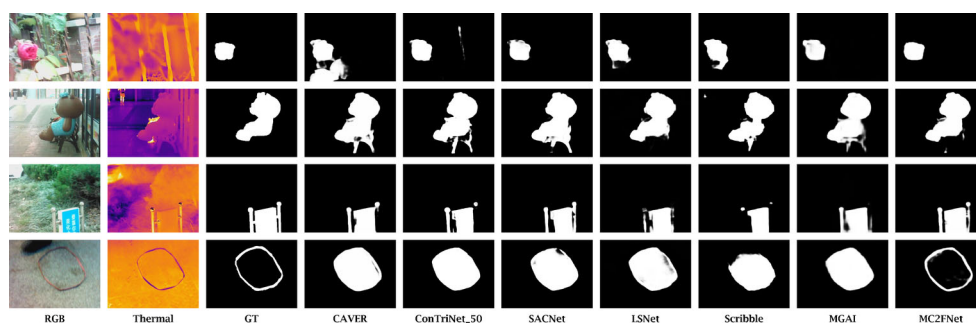


Fig. 11 Visual comparison examples between MC2FNet and SOTA methods on the VT5000 RGB-T dataset

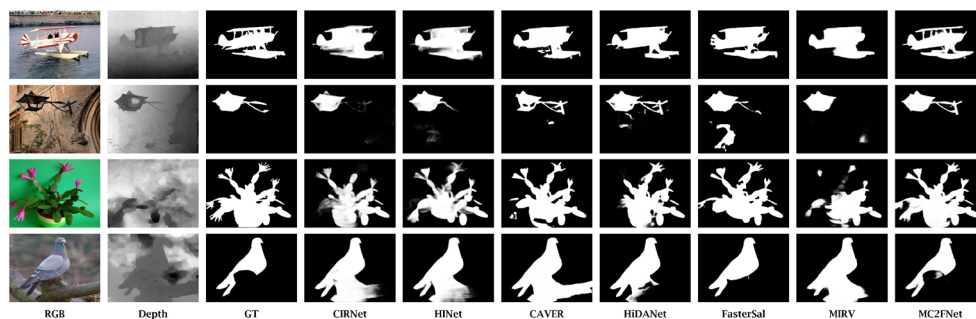


Fig. 12 Visual comparison examples between MC2FNet and SOTA methods on the NJUD RGB-D dataset

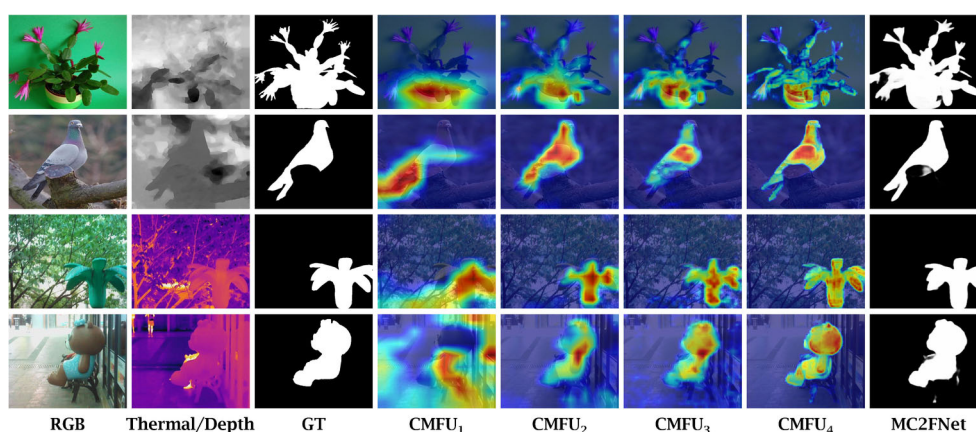


Fig. 13 Visualization of RGB-D/T salient object detection by MC2FNet. $CMFU_i$ represents the decoder for each stage

feature competitive fusion approach, on the other hand, can effectively complement ambiguous features, including those with similar color, depth, or thermal characteristics. Finally, the multi-scale competitive fusion approach leverages multi-level semantic information for cross-scale semantic complementation, offering a more effective alternative to direct element-wise summation.

In Fig. 13, we present a visual analysis of the output of the CMFU decoders at each level to further explore how different CMFUs process semantic information, providing deeper insights into their interpretability. MC2FNet exhibits robust detection performance for both single-target salient object detection and multi-target salient object detection.

According to the visualization results, the upsampling decoding process of CMFU progressively enhances the focus on salient objects, reduces structural-level errors down to the pixel level, and thereby contributes to achieving high performance on the E_{max}^{ξ} metric. When the target and background exhibit depth/thermal coherence or similar color and texture, CMFU enhances modal complementarity and promotes feature purification, thereby strengthening object perception and regional similarity assessment. This leads to a reduction in the model's MAE and an improvement in its S_m performance. Specifically, the output of $CMFU_1$ emphasizes global information, while $CMFU_2$ progressively shifts attention to salient objects and their surrounding regions. $CMFU_3$

Table 6 Proposed competitive fusion on the SIP dataset for a weighted ablation study. Best results are shown in Bold

Dataset	Intra-modal(IMCT)			Cross-modal(CMCT)			Multi-scale(DBIP)			MC2FNet $\alpha : \beta$
	$\alpha(0) : \beta(1)$	$\alpha(0.5) : \beta(0.5)$	$\alpha(1) : \beta(0)$	$\alpha(0) : \beta(1)$	$\alpha(0.5) : \beta(0.5)$	$\alpha(1) : \beta(0)$	$\alpha(0) : \beta(1)$	$\alpha(0.5) : \beta(0.5)$	$\alpha(1) : \beta(0)$	
SIP [62]	$S_m \uparrow$	0.8976	0.9003	0.9034	0.8986	0.9054	0.8953	0.9033	0.9016	0.9068
	$MAE \downarrow$	0.0409	0.0391	0.0383	0.0403	0.0368	0.0423	0.0372	0.0375	0.0359
	$max F \uparrow$	0.9091	0.9156	0.9149	0.9149	0.9190	0.9067	0.9144	0.9165	0.9200
	$E_{max}^{\xi} \uparrow$	0.9383	0.9420	0.9427	0.9387	0.9459	0.9341	0.9434	0.9437	0.9467

and CMFU₄ concentrate more directly on the salient objects themselves. This progression mirrors human visual attention, which transitions from a global search to a localized focus on salient objects.

4.3.2 Effective competitive fusion mechanism

We conducted ablation experiments on dynamic competitive adaptive fusion, and Table 6 shows that the proposed competitive fusion mechanism is effective on the SIP dataset. The high-quality fusion capability of intra-modal (IMCT), cross-modal (CMCT), and multi-scale (DBIP) competition is substantiated. We compare fixed-weight settings with competitively learned adaptive weights to evaluate the complementary feature fusion ability of CMFU.

The performance of the adaptive competitive learning factor for within-modal features is superior to fixed parameter settings, indicating that MC2FNet effectively captures feature variations across different spatial dimensions and achieves within-modal feature alignment through the hybrid strategy of GARSa and CRA. For cross-modal features, an adaptive competitive factor is used to select long-range complementary features, thereby facilitating efficient cross-modal fusion. Regarding multi-scale features, the adaptive competitive fusion of neighboring hierarchical features ensures semantic complementarity across hierarchical levels and promotes cross-hierarchical feature interactions at different granularities. As shown in Fig. 13, CMFU₂ enhances salient object detection and expedites the convergence of target objects through cross-layer interaction with CMFU₁.

Figure 14 illustrates the robustness advantages of the adaptive competitive fusion mechanism compared to fixed-weight fusion. The fixed-weight method struggles to segment salient objects when the target and background share similar colors, leading to a decline in $max F$ and E_{max}^{ξ} metrics. When the color or texture of the target varies dramatically, it fails to detect the target completely, which further impairs the S_m measure and increases MAE . Moreover, when the target and background are close in depth, its detection results show significant shortcomings, causing an overall degradation across multiple metrics. In contrast, MC2FNet addresses these issues effectively by simulating the neural enhancement and suppression processes in visual recognition through an adaptive competitive fusion mechanism and constructing a multi-scale cross-modal suppression–enhancement dynamic game model. As a result, it achieves overall improvements in key metrics such as MAE , S_m , E_{max}^{ξ} , and $max F$.

In practical data collection, image distortion frequently occurs. To validate the effectiveness of the competitive fusion mechanism under distorted data conditions, we collected experimental results from six groups of sensor-distorted samples. In Fig. 15, the distorted RGB modal data include cases such as image blur, overexposure, nocturnal data loss,

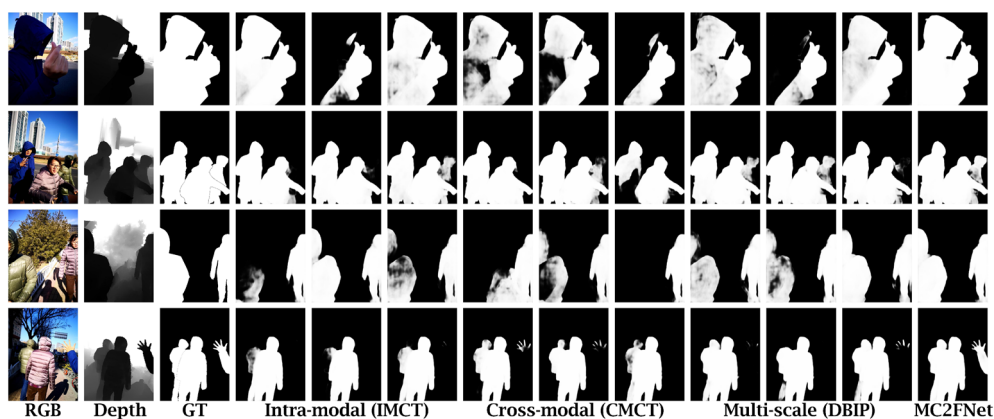


Fig. 14 Visual results of the weight ablation study of competitive fusion mechanisms on the SIP dataset. The visual results correspond to the results in Table 6 from left to right

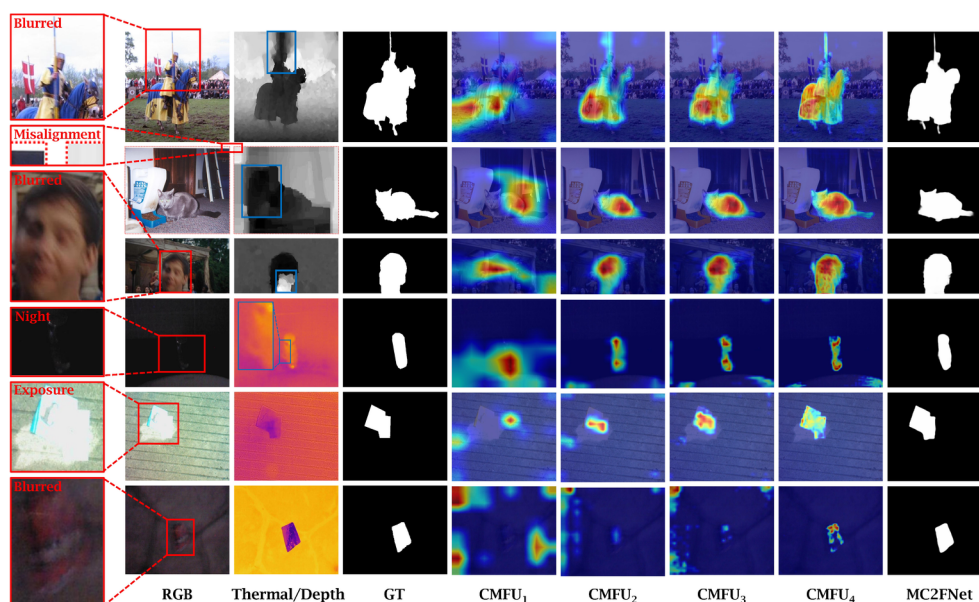


Fig. 15 Verification of MC2FNet's robustness under various data distortions. Red boxes indicate distortions in the RGB modal, and blue boxes indicate issues in the depth/thermal modal. $CMFU_i$ represents the decoder for each stage

and modal misalignment (marked with red boxes). The depth/thermal modality exhibits issues such as informational similarity or data distortion (indicated by blue boxes). The experiments confirm that, even under extreme distortion, the competitive fusion mechanism enables MC2FNet to consistently suppress MAE_2 , enhance S_m structural consistency, and maintain high E_{max}^{ξ} and $maxF$ scores. The CMFU attention maps from different decoding stages reflect how intra-modal, cross-modal, and multi-scale feature competition contributes to this robustness. Particularly in challenging scenarios like modal misalignment and nocturnal data loss (approaching modal absence), the model still achieves high-level SOD, which is directly reflected in its robust performance across MAE , S_m , E_{max}^{ξ} , and $maxF$ in Tables 1-2.

To further validate the robustness of MC2FNet under extreme conditions, we applied global masking to the depth/thermal infrared modality to simulate complete modality absence. The visual test results are shown in Fig. 16. As can be observed in the figure, when the depth/thermal infrared modality is fully masked (corresponding to the entirely black $Mask_{T/D}$ columns), the variant $MC2FNet_{mask}$ with masking constraints can still accomplish the object segmentation task relying solely on single-modality RGB information. However, compared to the original MC2FNet without masking, $MC2FNet_{mask}$ shows a decrease in segmentation performance, which further demonstrates that multimodal information can enhance the capability of salient object detection. It is worth affirming that even under the extreme condition of complete modality absence, $MC2FNet_{mask}$

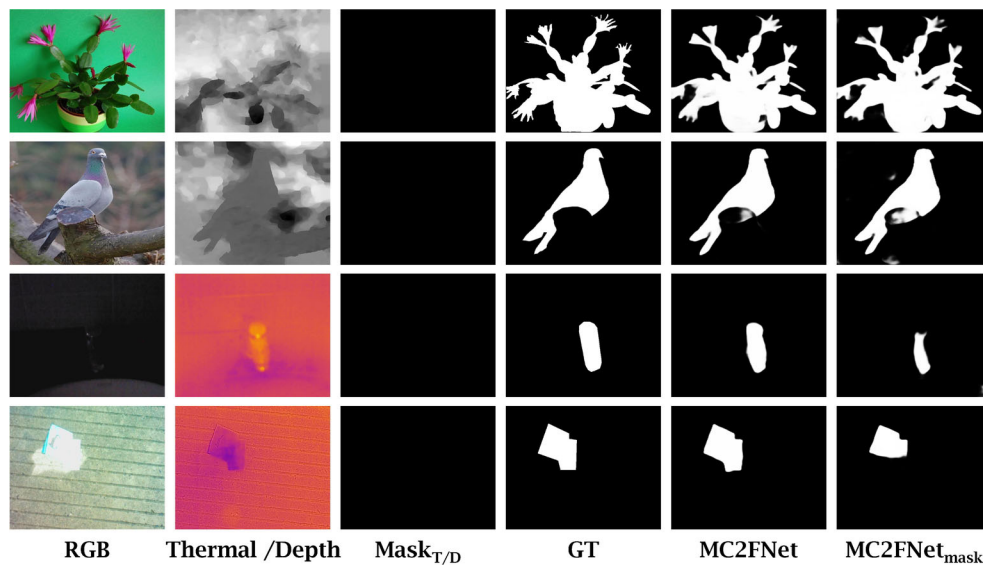


Fig. 16 Through complete masking of thermal and depth modal data, we simulate MC2FNet's testing performance under extreme modality-missing scenarios. The first column displays the RGB modality, the second column shows the thermal infrared or depth modality, and the third column represents the simulation of missing thermal infrared or

depth modality using full masking. The fourth column provides the ground truth, the fifth column presents the detection results of MC2FNet under normal multimodal conditions, and the sixth column shows the detection results of $MC2FNet_{mask}$ when thermal infrared or depth modality is simulated to be absent

still exhibits competitive performance. For example, in the first and second rows, even though the RGB modality provides limited effective information, the model can still detect the target object and achieve a high structural similarity (S_m). In the third and fourth rows, when the target in the RGB image resembles the background, the model attains an excellent E_{max}^s while maintaining strong detection capability. Overall, whether dealing with plants with complex branch structures, birds with clear contours, small targets under low visibility, or objects attached to walls, the model can generate segmentation masks that closely match the GT, demonstrating strong adaptability to scenarios with missing modalities. However, when the multimodal data provided at input is of suboptimal quality, MC2FNet currently struggles to detect entire salient objects with sufficient clarity.

Furthermore, we note that this experiment has certain limitations. Global black masking introduces an out-of-distribution input that does not fully reflect realistic modality degradation scenarios, such as sensor failure, partial occlusion, or low signal-to-noise conditions. Therefore, this demonstration should be regarded as a qualitative proof of concept for modality absence. To more rigorously evaluate the robustness of MC2FNet under missing modalities, future work should introduce both quantitative and qualitative analyses under real degradation scenarios. Nonetheless, the examples in Fig. 15 show that even under actual modality degradation conditions that cause low signal-to-noise ratios, such as overexposure or blurring, the model still maintains high detection performance. This indirectly demonstrates,

to some extent, the robustness of MC2FNet under modality absence.

4.3.3 Effective simplified attention

To evaluate the inference performance of the two simplified attention mechanisms, we measured the model inference speed under different parameter settings. As shown in Table 7, the inference speed varies across different attention mapping strategies, and different combinations of GARS/GARSCA and CRA/CRCA also lead to differences in computational efficiency. When CRA/CRCA uses the full channel count and the embedding dimension of GARS/GARSCA is not reduced, the model attains the highest computational cost at 142.4 GFLOPs. In contrast, when CRA/CRCA is reduced to a single channel and each layer of GARS/GARSCA uses only a single token, the computational cost is minimized at 50.4 GFLOPs. In the actual training of MC2FNet, with the attention parameters set as indicated in Table 7, the inference cost reaches 50.7 GFLOPs. These results under various parameter configurations demonstrate that MC2FNet significantly improves inference efficiency and achieves high-quality multimodal salient object detection with low computational complexity.

We decomposed the core modules of each level of the decoder's CMFU, namely IMCT, CMCT, DBIP, and CST, and calculated the computational costs of these corresponding core modules, demonstrating the efficiency of the two simplified attention mechanisms. As shown in Table 8, as the

Table 7 Complexity comparison of the number of embedded tokens in different layers of GARS/GARSCA (G) and CRA/CRCA (C). For example, column 3 is the re-embedding of 2^2 tokens into 1 token for both different tiers, and the channel is reduced to 1

Metrics	G:(1,1,1,1) C:Reduce to 1	G:(2,2,2,2) C:Reduce to 1	G:(4,4,4,4) C:Reduce to 1	G:(8,16,32,64) C:Reduce to 1	G:(8,8,8,8) C:Reduce by 1/2	G:(8,8,8,8) C:No reduce	G:(4,4,4,4) C:No reduce	G:(2,2,2,2) C:No reduce	G:(1,1,1,1) C:No reduce	G:(8,8,8,8) C:Reduce to 1
FLOPs(G)↓	110.1	65.4	54.2	50.4	51.4	83.8	86.6	97.8	142.4	50.7(MC2FNet)

Table 8 Break down the core modules of IMCT, CMCT, DBIP, and CST across each decoding level of CMFU, and test the computational cost (GFLOPs) of the corresponding core modules

Decoder Layer	Intra-modal (IMCT)	Cross-modal (CMCT)	Multi-scale (DBIP)	Cross-Scale intra-modal (CST)
$CMFU_1$	0.02	0.03	-	0.08
$CMFU_2$	0.08	0.11	0.58	0.32
$CMFU_3$	0.33	0.44	2.31	1.28
$CMFU_4$	1.32	1.76	9.23	5.11

decoder level deepens progressively, the computational costs (GFLOPs) of each core module show an increasing trend. The computational cost of IMCT rises from 0.02 GFLOPs to 1.32 GFLOPs, CMCT from 0.03 GFLOPs to 1.76 GFLOPs, DBIP from 0.58 GFLOPs to 9.23 GFLOPs, and CST from 0.08 GFLOPs to 5.11 GFLOPs. Among these, DBIP has the highest computational cost because it needs to process the concatenated features of two modalities and also includes IMCT. This quantified variation in computational cost clearly reveals the differences in computational load across modules at different levels. The two simplified attention mechanisms optimize the model computation, effectively controlling the increase in computational cost, and achieve better efficiency performance within the complex multimodal decoding process.

Table 9 compares the complexity of MC2FNet with recent SOTA RGB-T/D salient object detection methods. MC2FNet achieves the best overall performance and the highest computational efficiency across both modalities. MC2FNet has 107.7M parameters, which is in the moderate range. It is lower than SACNet (327.7M) and higher than ConTriNet50 (34.8M) and SPSN (37.0M), which are in the lowest parameter range. Moreover, the use of different pre-trained models across methods is one of the reasons for the difference in parameter counts. RD3D has the highest inference computation (508.6 GFLOPs), while MC2FNet requires only 50.7 GFLOPs, approximately one-tenth of that of RD3D. This parameter configuration is designed to ensure sufficient modeling capability for complex scenes, laying the foundation for SOTA detection performance. Based on the efficiency analysis on the VT1000 and NJUD datasets, the parameter efficiency of MC2FNet (S_m /Params of 0.0087) is close to that of HINet (0.0093) in the same parameter scale and is higher than that of SACNet (0.0062), indicating no redundancy in parameter count. In practical deployment, inference computation is more critical than parameter count. MC2FNet achieves the highest computational efficiency (S_m /FLOPs of 0.0186) in RGB-T scenarios, significantly outperforming all compared methods. These results fully validate the effectiveness of the two attention mixing strategies in CMFU. They

Table 9 Comparison of the complexity of some recent publicly available state-of-the-art RGB-T/D SOD methods

Metrics	CVPR20 S2MA [74]	ECCV20 CMWNet [27]	TMM20 ADF [56]	TIP21 MIDD [66]	AAAI21 RD3D [76]	ECCV22 SPSN [78]	TCSVT22 CGFNet [46]	TCSVT22 MGAI [71]	PR23 HINet [79]	TMM24 SACNet [12]	TPAMI25 ConTriNet50 [52]	MC2FNet Ours
Params(M) ↓	86.6	70.9	83.1	52.4	46.9	37.0	69.9	89.0	98.9	327.7	34.8	107.7
FLOPs(G) ↓	141.1	482.3	191.5	216.7	508.6	100.3	347.8	78.2	58.3	149.9	55.4	50.7
S_m /Params ↑	0.0125	0.0127	0.0109	0.0173	0.0195	0.0248	0.0132	0.0320	0.0093	0.0028	0.0267	0.0087
S_m /FLOPs ↑	0.0063	0.0019	0.0047	0.0042	0.0018	0.0092	0.0027	0.0119	0.0157	0.0062	0.0168	0.0186

successfully decouple parameter count from inference computation, reducing redundant floating-point operations while maintaining sufficient modeling capacity, thus achieving an optimal balance between detection performance and inference efficiency.

Hybrid attention mechanisms with different parameter combinations lead to varying performance outcomes, as validated on the RGB-T SOD datasets VT821 and VT1000. As shown in Table 10, the parameter configuration used in MC2FNet achieves the best performance in most cases. It is noteworthy that the model still attains the second-best performance even when the per-layer mapping parameter of GARS/GARSCA is set to 4, and CRA/CRCA reduces the number of channels to 1. This indirectly demonstrates the effectiveness of employing different embedding mapping strategies to capture feature distinctions. The experimental results indicate the presence of redundant features in attention computation. The hybrid strategy of mapping the two attention mechanisms to different spatial dimensions effectively captures feature differences and mitigates the interference caused by redundant features during extraction, underscoring the importance of spatial structure. Furthermore, the results in Table 10 confirm the efficacy of the competitive fusion mechanism, which dynamically balances suppression and enhancement within the spatial structure and enables fine-grained attention modulation through competition across hierarchical scales, thereby promoting feature purification.

Figure 17 presents visual results under different embedding mapping parameters listed in Table 10. The results demonstrate that MC2FNet effectively suppresses the influence of redundant features on detection outcomes in both simple and complex environments. In particular, with its selected parameters, MC2FNet achieves optimal detection performance in challenging scenarios, such as those with color or material similarity, or target-background thermal consistency, which is reflected in superior scores on metrics including E_{max}^{ξ} , $max F$, and S_m , along with low MAE values, while maintaining low computational complexity. This advantage stems from its two simplified attention mechanisms, which extract discriminative spatial features by projecting modalities into different dimensions to capture inter-modal differences, thereby enhancing structural consistency (S_m), improving overall accuracy (E_{max}^{ξ} , $max F$), and reducing pixel-wise error (MAE).

5 Discussion

5.1 Advantages

MC2FNet significantly improves the accuracy of multimodal salient object detection by emulating the spatial synergistic suppression–enhancement mechanism in visual cognition.

Table 10 Comparison of model performance between GARS/GARSCA and CRA/CRA for different embedding mapping parameters on two RGB-T SOD datasets

Models			VT821				VT1000			
			$S_m \uparrow$	$MAE \downarrow$	$max F \uparrow$	$E_{max}^s \uparrow$	$S_m \uparrow$	$MAE \downarrow$	$max F \uparrow$	$E_{max}^s \uparrow$
A	GARS/GARSCA CRA/CRA	G:(2,2,2,2) C:Reduce to 1	0.8933	0.0288	0.8725	0.9285	0.9384	0.0173	0.9392	0.9717
B	GARS/GARSCA CRA/CRA	G:(4,4,4,4) C:Reduce to 1	0.8980	0.0273	0.8800	0.9386	0.9407	0.0159	0.9402	0.9736
C	GARS/GARSCA CRA/CRA	G:(8,16,32,64) C:Reduce to 1	0.8912	0.0268	0.8629	0.9253	0.9390	0.0162	0.9376	0.9707
D	GARS/GARSCA CRA/CRA	G:(8,8,8,8) C:Reduce by 1/2	0.8868	0.0300	0.8608	0.9258	0.9398	0.0163	0.9395	0.9730
E	GARS/GARSCA CRA/CRA	G:(8,8,8,8) C:Reduce by 1/4	0.8876	0.0292	0.8611	0.9198	0.9383	0.0165	0.9397	0.9721
F	MC2FNet	G:(8,8,8,8) C:Reduce to 1	0.8952	0.0268	0.8746	0.9301	0.9422	0.0153	0.9435	0.9753

The top-ranked results are highlighted in Bold, while the second-ranked ones are highlighted in Italic

Its core competitive advantage lies in the hierarchical feature extraction architecture and cross-modal competitive fusion strategy, which effectively addresses detection failures caused by target-background depth/thermal consistency or color-texture similarity. The CMFU module integrates a hybrid attention mechanism combining Channel Reduction (CR) and Group Aggregation Token Re-embedding (GATR), achieving efficient integration of intra-modal, cross-modal, and multi-scale features at low computational complexity. By simulating neural enhancement and suppression in spatial structures through dynamic adaptive competitive modeling, the method enhances the complementary capabilities of multimodal information, particularly in complex scenarios, demonstrating superior semantic consistency and noise suppression. Experimental results confirm that MC2FNet achieves state-of-the-art performance in multimodal salient object detection tasks.

5.2 Limitations and future work

Although MC2FNet achieves notable advances in both performance and efficiency, several aspects still require further investigation. The current model has been primarily validated on RGB-Depth and RGB-Thermal data, and its generalizability to other multimodal combinations remains to be thoroughly evaluated. Moreover, despite achieving end-to-end learnable competitive weighting and substantially reduced computational complexity, the model's robustness under extreme real-world conditions, such as significant occlusion or missing modalities, needs additional study. While target objects can be detected under simulated modal missing conditions, detection performance deteriorates when multimodal input data quality is suboptimal, indicating room for further improvement. Future work will focus on extending the model's applicability to a broader range of multimodal settings, including more demanding applications such as mobile robotic perception and autonomous driving. We will also explore integrating newly discovered competitive regulatory mechanisms from neuroscience to improve the model's interpretability and adaptability.

6 Conclusion

In this paper, we revisit the architecture of multimodal salient object detection (SOD) from the perspective of intra-modal, cross-modal, and multi-scale competitive fusion, and propose a novel spatially-aware SOD network named MC2FNet. Inspired by the spatial synergistic suppression-enhancement mechanism, MC2FNet mimics the visual system's feature refinement process from coarse to fine. The network employs a specially designed CMFU module for efficient cross-modal feature fusion and decoding and incorporates two attention

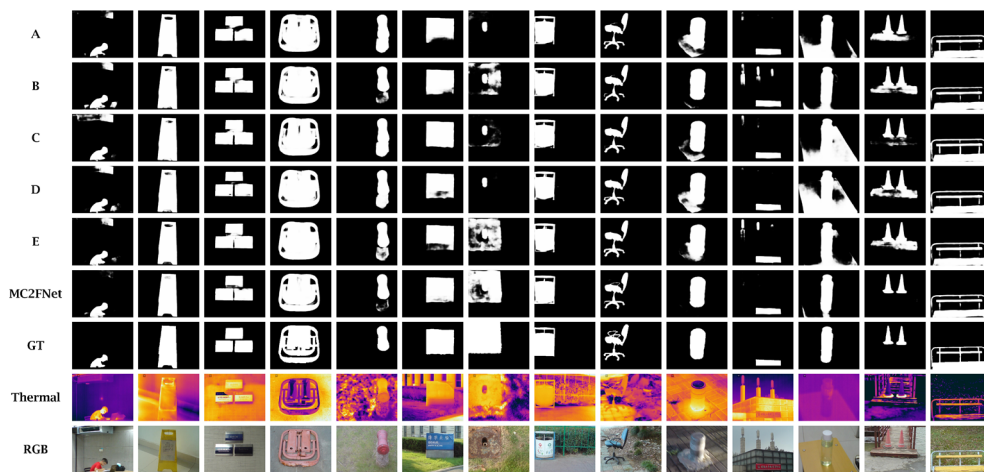


Fig. 17 Comparison of visual results of GARSA/GARSCA and CRA/CRCA for different embedding mapping parameters on the VT1000 RGB-T SOD dataset. The visual results correspond to the results in Table 10 from top to bottom

mechanisms, namely CRA and GARSA, to reduce computational complexity while maintaining high feature extraction performance, making it suitable for real-world SOD applications. Furthermore, the DBIP module leverages multi-scale semantic features to accomplish cross-scale feature fusion. Extensive experiments validate the effectiveness of the proposed method.

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Author Contributions H.T. conceived and designed the experiments; H.T. and S.W. performed the experiments and organized the experimental data; H.T., S.W., T.H., and L.Z. analyzed the results and drafted the manuscript; T.H. and J.L. provided suggestions; and all authors reviewed and edited the manuscript.

Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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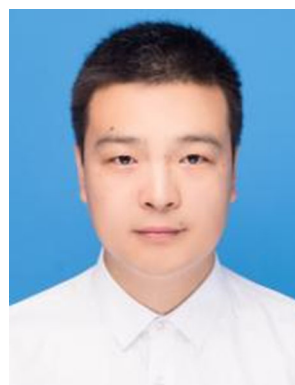
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